Token-Level Uncertainty as a Predictor of Hallucination in Large Language Models

Abstract

Large language models (LLMs) routinely generate fluent, syntactically well-formed text that is nonetheless factually incorrect or unsupported by evidence — a phenomenon widely termed “hallucination.” Because hallucinations are often stylistically indistinguishable from correct output, detecting them post hoc is difficult, and many proposed solutions instead look *inward*, using signals available during generation itself. Among these, token-level uncertainty — quantities such as next-token entropy, log-probability, and perplexity computed at each decoding step — has emerged as one of the most widely studied predictors of hallucination, on the intuition that a model “guesses” when it is uncertain. This paper surveys the theoretical basis, empirical evidence, and current methodological landscape connecting token-level uncertainty to hallucination in LLMs. We review foundational findings linking predictive uncertainty and hallucination in conditional generation, the extension of these ideas to autoregressive LLMs through log-probability and entropy-based scores, and their evolution into semantic-level and hybrid approaches such as semantic entropy, semantic entropy probes, and attention-weighted uncertainty measures. We then examine current systems that operationalize token-level uncertainty for hallucination detection, fact-checking, and abstention, alongside evidence on calibration and its limits. We identify persistent challenges — miscalibration and overconfidence, the confound between epistemic and aleatoric uncertainty, lexical variation masking semantic agreement, computational overhead of sampling-based methods, and the structural incentive within training and evaluation pipelines for models to guess rather than express uncertainty. We conclude by outlining research gaps, including the need for uncertainty measures that generalize across black-box and white-box settings, better theoretical grounding for token-to-claim aggregation, and evaluation protocols that isolate the causal role of uncertainty in hallucination rather than mere correlation.

Keywords: large language models; hallucination detection; token-level uncertainty; predictive entropy; semantic entropy; calibration; natural language generation; uncertainty quantification

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1. Introduction

Large language models (LLMs) such as GPT-4, Gemini, and Claude have achieved remarkable fluency and apparent competence across open-ended generation tasks, from question answering to summarization to multi-step reasoning. Yet this fluency conceals a persistent weakness: LLMs frequently produce statements that are plausible in form but false, unsupported, or fabricated in content — a behavior the field has come to call “hallucination” (Ji et al., 2023). Hallucinations are not merely cosmetic errors. They have been linked to the fabrication of legal citations, false claims in generated news content, and risks in safety-critical domains such as medical diagnosis support (Farquhar et al., 2024). Because hallucinated text is often fluent and internally coherent, human readers — and even automated fact-checking pipelines relying on surface plausibility — can struggle to distinguish it from accurate content, which amplifies its practical danger relative to more obviously anomalous errors.

A central research question, then, is whether hallucinations can be anticipated or flagged using signals internal to the generation process itself, rather than relying solely on post hoc verification against external knowledge sources. Token-level uncertainty — measures such as the probability assigned by the model to the token it generates, the entropy of the next-token distribution, or the log-probability of a generated sequence — is a natural candidate. The underlying hypothesis, articulated across a range of studies, is that a model is more likely to "guess" or generate an unsupported claim precisely when its internal predictive distribution is diffuse or when it assigns low probability to the tokens it ultimately emits (Xiao & Wang, 2021; Malinin & Gales, 2021). This intuition connects hallucination detection to a much older body of work on predictive uncertainty and calibration in machine learning (Guo et al., 2017), while also raising LLM-specific complications: language admits many surface forms for the same meaning, uncertainty about *phrasing* is not the same as uncertainty about *facts*, and modern LLMs are known to be poorly calibrated, particularly after instruction tuning and reinforcement learning from human feedback (Kadavath et al., 2022).

This paper reviews the state of research on token-level uncertainty as a predictor of hallucination in LLMs. Section 2 provides background on hallucination taxonomy and the theoretical motivation for uncertainty-based detection. Section 3 offers a technical discussion of the principal families of token-level uncertainty measures and how they relate to hallucination. Section 4 surveys current approaches and systems that operationalize these measures, including sampling-based, single-pass, and hybrid semantic methods. Section 5 discusses methodological and practical limitations. Section 6 identifies open research gaps and promising directions. Section 7 concludes.

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2. Background and Related Work

2.1 Defining and Categorizing Hallucination

Hallucination in natural language generation (NLG) predates the LLM era, having been studied extensively in machine translation, summarization, and data-to-text generation (Ji et al., 2023). Ji et al. (2023) offer a widely cited taxonomy distinguishing *intrinsic* hallucinations — output that directly contradicts the source or input content — from *extrinsic* hallucinations, where the output includes content that cannot be verified from the source at all, whether or not it happens to be true. In the LLM setting, where there is frequently no single "source" document constraining the generation (as in open-domain question answering or free-form dialogue), the operative notion of hallucination shifts toward *factuality*: whether the generated claim is true and verifiable against reality or an authoritative external knowledge base (Farquhar et al., 2024).

Large-scale surveys of LLM hallucination (Ji et al., 2023; Huang et al., 2025) converge on several contributing factors: gaps or errors in pretraining data, exposure bias and error accumulation during autoregressive decoding, insufficient grounding to retrieved or provided context, and misaligned training objectives that reward confident, complete-sounding answers over calibrated expressions of uncertainty. This last factor has received particular theoretical attention. Kalai et al. (2025) argue that hallucinations arise as a natural consequence of treating next-token prediction as a form of binary classification over plausible versus implausible continuations: if incorrect statements are not statistically distinguishable from correct ones in the training distribution, some hallucination is a mathematically inevitable byproduct of the pretraining objective, and is subsequently reinforced by evaluation and benchmark procedures that penalize abstention ("I don't know") more heavily than confident wrong answers (Kalai et al., 2025). This account is important context for the present review, because it implies that low predictive uncertainty is not always trustworthy evidence of correctness — a model can be confidently wrong precisely because its training incentivized confidence over honesty.

2.2 Predictive Uncertainty as a Signal: Origins

The link between predictive uncertainty and hallucination was established prior to the emergence of modern LLMs. Xiao and Wang (2021) systematically studied hallucination in conditional generation tasks (image captioning and data-to-text generation) and found that higher predictive uncertainty — decomposed into epistemic and aleatoric components — correlates with a higher incidence of hallucination, with epistemic uncertainty (uncertainty due to limited model knowledge, as opposed to inherent ambiguity in the task) being the more informative of the two (Xiao & Wang, 2021). They further proposed an uncertainty-aware extension to beam search that reduced hallucination while preserving task performance, providing early causal-adjacent evidence (via intervention on the decoding procedure) that uncertainty signals are not merely correlated with, but exploitable for reducing, hallucination.

Malinin and Gales (2021) generalized uncertainty estimation to autoregressive structured prediction more broadly, formalizing uncertainty measures at both the token level and the full-sequence level within an ensemble-based probabilistic framework, and distinguishing measures that capture "knowledge uncertainty" (epistemic) from those capturing inherent data ambiguity (Malinin & Gales, 2021). This work supplied much of the formal vocabulary — entropy of the token distribution, mutual information between model parameters and predictions, and sequence-level log-probability — that later LLM-hallucination research adapted directly.

2.3 Calibration as a Prerequisite

A necessary condition for token-level uncertainty to be a *useful* hallucination predictor is that the model's probabilities be reasonably calibrated — that is, that a token or sequence assigned probability p is correct with frequency approximately p. Guo et al. (2017)

demonstrated that modern deep neural networks, despite improved accuracy relative to older architectures, tend to be poorly calibrated and systematically overconfident, and popularized temperature scaling as a simple, effective post hoc correction (Guo et al., 2017). Kadavath et al. (2022) extended calibration analysis specifically to LLMs, showing that large models are relatively well calibrated on diverse multiple-choice and true/false formats when queried in the right format, and that models can be trained to directly predict "P(True)" — the probability that a proposed answer is correct — often outperforming naive use of the model's own per-token probabilities, especially in open-ended generation settings where next-token probability conflates uncertainty about *content* with uncertainty about *phrasing* (Kadavath et al., 2022). This distinction — between uncertainty over what to say and uncertainty over how to say it — is the central motivation for the semantic-level uncertainty measures discussed in Section 3.

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3. Main Technical Discussion: Token-Level Uncertainty Measures and Their Relationship to Hallucination

3.1 Basic Token-Level Signals

The simplest and most widely used token-level uncertainty signals derive directly from the model's output logits at each decoding step:

- **Token probability / negative log-likelihood**: the probability the model assigns to the token it ultimately generates, or equivalently its negative log-probability, at each position.
- **Sequence log-probability**: the (length-normalized) sum of per-token log-probabilities across a generated span, used by Guerreiro et al. to detect hallucination in machine translation by measuring the model's confidence in its own output as a whole (as reported in Alkaissi & Lee's survey lineage cited in Huang et al., 2025 and detailed in the comprehensive survey by Huang et al., 2025, drawing on Guerreiro et al.'s sequence log-probability approach).
- **Predictive entropy**: the Shannon entropy of the next-token probability distribution at a given position, $H_t = -\sum_{v \in V} p\,(v \mid x_{<t}) \log p(v \mid x_{<t})$, which is high when the model spreads probability mass across many plausible continuations and low when it is sharply peaked on a single token.
- **Perplexity**: the exponentiated average negative log-likelihood over a generated sequence, functioning as an aggregate, sequence-level uncertainty score.

The theoretical hypothesis motivating these measures is straightforward: a hallucinated token, by definition, is one the model is fabricating rather than retrieving from reliably learned knowledge, and fabrication should on average correspond to a flatter, more uncertain next-token distribution — the model is "guessing instead of relying on learned patterns" (Huang et al., 2025). This intuition has been empirically supported across several studies. Chuang et al. and Das et al. found that token-level probability distribution entropy correlates with factual hallucinations, with models exhibiting higher entropy when generating factually incorrect content (as synthesized in the retrieval-augmented generation study of context-faithfulness hallucination; see Huang et al.'s DAGCD analysis, arXiv:2501.01059). Similarly, in the fact-checking pipeline of Fadeeva et al. (2024), token-level uncertainty quantification was used directly to fact-check atomic claims extracted from LLM output, isolating uncertainty attributable to *what claim to make* from uncertainty attributable to *surface realization* — an important refinement that improves the diagnostic value of token-level signals (Fadeeva et al., 2024).

3.2 Sequence-Level Aggregation and Its Difficulties

A practical challenge is that hallucinations are rarely confined to a single token; they manifest as entities, relations, or entire claims spanning multiple tokens, and the "most uncertain token" in a hallucinated span may not itself carry the erroneous content (e.g., a function word rather than the fabricated entity name). Naively averaging token probabilities across a span can dilute the signal, since function words are typically predicted with high confidence regardless of the veracity of the surrounding content. This has motivated approaches that reweight token-level uncertainty by informativeness. Zhang et al. (2023) enhanced uncertainty-based hallucination detection by improving token probability estimation, reducing overconfidence, and introducing a focus mechanism that emphasizes the most informative keywords and historically unreliable tokens rather than treating all tokens uniformly (Huang et al., 2025, summarizing Zhang et al., 2023). Duan et al. (2024) similarly proposed "Shifting Attention to Relevance" (SAR), reassigning attention toward semantically relevant tokens and components when computing uncertainty, rather than weighting all tokens in a sequence equally (Duan et al., 2024, arXiv:2307.01379).

3.3 From Lexical to Semantic Uncertainty

A further complication specific to natural language is that the same underlying claim can be expressed in many different surface forms (e.g., "Paris is the capital of France" versus "France's capital is Paris"), such that lexical variability across sampled generations may resemble uncertainty even when the model is, in fact, consistently expressing the same — possibly correct — belief. Purely token-level, lexically grounded uncertainty measures are blind to this distinction and can register high uncertainty for stylistic reasons alone (Huang et al., 2025).

This limitation motivated the development of *semantic* uncertainty measures. Kuhn, Gal, and Farquhar (2023) introduced semantic entropy: multiple candidate generations for the same prompt are sampled, clustered into equivalence classes based on bidirectional entailment (using a natural language inference, or NLI, model to determine whether two generations express the same meaning), and entropy is computed over the resulting *cluster* distribution rather than over raw token sequences (Kuhn et al., 2023). Farquhar, Kossen, Kuhn, and Gal (2024) extended and scaled this method, demonstrating in a *Nature* publication that semantic entropy is a reliable, unsupervised, and architecture-agnostic signal for detecting confabulations — a specific subtype of hallucination characterized by arbitrary and inconsistent wrong answers — across question-answering and other free-form generation tasks, without requiring task-specific supervision (Farquhar et al., 2024). This work is significant for the present discussion because it demonstrates that semantic entropy, while computed from the same underlying token-level probabilities used to sample multiple generations, resolves the lexical-confound problem inherent in naive token-level entropy by operating at the level of meaning rather than surface form.

Because sampling many full generations per query and running NLI-based clustering is computationally expensive, subsequent work has sought cheaper approximations. Kossen et al. (2024) proposed Semantic Entropy Probes (SEPs), which approximate semantic entropy directly from a single forward pass's hidden states via lightweight linear probes, substantially reducing the five- to ten-times inference overhead associated with the sampling-based original method while retaining much of its predictive value (Kossen et al., 2024, arXiv:2406.15927). Related extensions include Kernel Language Entropy (Nikitin et al.), which generalizes the clustering step to a continuous kernel-based similarity measure over sampled generations rather than discrete entailment clusters, yielding finer-grained uncertainty quantification (as cited in Kossen et al., 2024, and subsequent literature).

3.4 Variance- and Consistency-Based Approaches

A parallel line of work sidesteps explicit entropy computation in favor of measuring the *variance* or *consistency* of model behavior across repeated stochastic generations. SelfCheckGPT (Manakul, Liusie, & Gales, 2023) operationalizes the idea that if a model genuinely "knows" a fact, independently sampled generations about that fact should agree with one another, whereas hallucinated content will tend to vary and contradict itself across samples; the method requires no access to model-internal probabilities and thus works even in fully black-box settings, using surface-level consistency metrics such as BERTScore, NLI agreement, and LLM-based comparison prompts (Manakul et al., 2023). Detecting Token-Level Hallucinations Using Variance Signals extends this consistency intuition to the token level, flagging individual tokens as likely hallucinated when their log-probabilities exhibit high variance across multiple stochastic regenerations of the same context, offering a reference-free, model-agnostic, and interpretable localization of hallucinated spans within otherwise fluent text (arXiv:2507.04137). These variance-based methods are conceptually continuous with token-level uncertainty in that both rely on the instability of the model's predictive distribution as the diagnostic signal, but they substitute *empirical* variance across samples for the *analytic* entropy of a single distribution, trading computational efficiency (single forward pass) for robustness to certain forms of overconfidence.

3.5 Internal-State and Hybrid Approaches

Beyond output-layer probabilities, several approaches probe hidden representations directly, on the premise that a model's internal states may encode uncertainty or truthfulness signals not fully reflected in the output distribution. Azaria and Mitchell (2023) showed that internal activations of an LLM can reveal whether the model is generating a statement it "believes" to be false, and Chen et al. (2024, INSIDE) similarly used internal states to retain hallucination-detection information beyond what is available from output probabilities alone. Zhang and colleagues have proposed detection frameworks classifying uncertainty-based hallucination methods into token-based, semantic-based, and structured/supervised categories, with hybrid approaches increasingly combining logit-derived uncertainty with hidden-state probes and, in retrieval-augmented settings, attention-based signals over the retrieved context (Chuang et al., 2024; Huang et al., 2025). For example, work on context-faithfulness hallucination in retrieval-augmented generation has shown that token-level probability distribution entropy correlates with whether a model's output is faithful to retrieved evidence, motivating decoding-time interventions such as dynamic attention-guided context decoding that use entropy signals to adjust the influence of retrieved context during generation (arXiv:2501.01059).

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4. Current Approaches and Developments

The research reviewed above has translated into several classes of practical hallucination-detection and mitigation systems that operationalize token-level uncertainty:

Fact-checking pipelines. Fadeeva et al. (2024) present a pipeline that uses token-level uncertainty quantification to fact-check atomic claims extracted from LLM output, explicitly separating uncertainty about *claim selection* from uncertainty about *surface realization*, enabling more precise localization of unreliable spans within otherwise fluent generations (Fadeeva et al., 2024).

Scalable and reference-free detectors. TOKENHD (Min, Pang, Du, Cheng, & Fung) trains dedicated token-level hallucination detectors using a scalable synthetic data engine and an importance-weighted training recipe, showing that even compact detectors (0.6B parameters) can outperform much larger reasoning models at localizing hallucinated tokens in free-form text, without requiring predefined step segmentation (arXiv:2605.12384). Variance-signal approaches such as the reference-free token-level detector described in Section 3.4 similarly aim for scalability and applicability without ground-truth labels or external knowledge bases (arXiv:2507.04137).

Density- and proxy-based uncertainty for truthfulness. Token-level density-based uncertainty quantification methods estimate the density of a generated token's representation relative to the model's training distribution as an additional signal correlated with truthfulness, complementing raw probability-based measures (arXiv:2502.14427).

Reasoning-specific uncertainty estimation. TokUR (Token-Level Uncertainty Estimation for LLM Reasoning) addresses the reasoning setting specifically, where multi-step chains of thought can accumulate hallucinated intermediate steps that are difficult to detect at the level of a single final answer; token-level uncertainty is aggregated across reasoning steps to flag unreliable intermediate conclusions (arXiv:2505.11737).

Calibration and abstention training. Building on the diagnosis that current training and evaluation incentives reward confident guessing over calibrated abstention (Kalai et al., 2025), a growing set of approaches attempts to directly train models to express verbal or probabilistic uncertainty that better reflects their internal token-level confidence, including calibrating verbal uncertainty as a linear feature and confidence-aware abstention mechanisms that use internal uncertainty signals to decide when a model should decline to answer rather than produce a hallucinated response.

Collectively, these systems illustrate a shift from treating token-level uncertainty as a passive diagnostic (used only in post hoc detection) to an *active* signal incorporated into decoding (e.g., uncertainty-aware beam search, attention-guided decoding), training (calibration-aware fine-tuning), and interactive abstention mechanisms.

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5. Research Challenges and Limitations

Despite substantial progress, several challenges limit the reliability and generality of token-level uncertainty as a hallucination predictor.

Miscalibration and overconfidence. Instruction tuning and reinforcement learning from human feedback, while improving helpfulness and coherence, have been repeatedly observed to degrade the calibration of raw output probabilities relative to base pretrained models, weakening the direct correspondence between token probability and correctness that uncertainty-based methods rely upon (Kadavath et al., 2022; Guo et al., 2017). A model can be confidently wrong precisely because training incentives reward assertive, complete-sounding answers over calibrated hedging (Kalai et al., 2025), which undermines the core assumption that low uncertainty implies reliability.

Epistemic versus aleatoric confound. Raw entropy or log-probability measures conflate uncertainty arising from genuine gaps in model knowledge (epistemic uncertainty, which is diagnostic of hallucination risk) with uncertainty arising from inherent ambiguity or multiple valid phrasings of a correct answer (aleatoric uncertainty, which is not) (Malinin & Gales, 2021; Xiao & Wang, 2021). Disentangling these components typically requires ensembles, repeated sampling, or auxiliary models, adding computational cost.

Lexical variation masking semantic agreement. Because natural language permits many surface realizations of the same meaning, naive token- or sequence-level probability comparisons across samples can register spurious "uncertainty" driven by paraphrase variability rather than factual unreliability, a limitation explicitly motivating the shift to semantic-level clustering methods (Kuhn et al., 2023; Farquhar et al., 2024; Huang et al., 2025).

Computational overhead. The most robust semantic and consistency-based methods (semantic entropy, SelfCheckGPT, variance-based sampling detectors) require multiple stochastic generations per query, multiplying inference cost by a factor often cited as five- to ten-fold or more relative to greedy decoding, which is a substantial barrier to real-time or large-scale deployment (Kossen et al., 2024; Manakul et al., 2023). Cheaper single-pass approximations (e.g., semantic entropy probes, internal-state probes) trade some predictive fidelity for efficiency, and the extent of this trade-off is not yet fully characterized across domains and model families.

Black-box inaccessibility. Many of the most informative token-level signals (full output distributions, hidden states) are unavailable for proprietary, API-only models, restricting uncertainty-based detection in commercial deployment settings to either sampling-based, black-box-compatible methods (e.g., SelfCheckGPT) or to coarser API-exposed signals such as top-k log-probabilities, when these are offered at all (Manakul et al., 2023).

Aggregation from token to claim. There is no settled, theoretically justified method for aggregating token-level uncertainty scores into claim-level or passage-level hallucination judgments; simple averaging, minimum, or maximum aggregation across a span each carry different, sometimes counterintuitive, failure modes, particularly when hallucinated content is concentrated in a small number of highly confident but incorrect tokens (Huang et al., 2025).

Task and domain generalization. Much of the empirical evidence for uncertainty-hallucination correlations comes from question answering, summarization, image captioning, and machine translation benchmarks; generalization to long-form generation, multi-turn dialogue, and complex multi-step reasoning tasks — where hallucinations can arise and compound at any of many intermediate steps — remains comparatively understudied, though recent work on reasoning-specific token uncertainty (TokUR) and reasoning-hallucination detectors (TOKENHD) represents early progress in this direction.

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6. Research Gaps and Future Directions

Building on the challenges above, we identify the following priority directions for future research.

Causal, not merely correlational, evaluation. The bulk of existing evidence establishes correlation between token-level uncertainty and hallucination incidence; comparatively fewer studies causally intervene on uncertainty-related quantities during decoding (an exception being the uncertainty-aware beam search of Xiao & Wang, 2021) to test whether reducing measured uncertainty at generation time actually reduces hallucination rates, as opposed to merely correlating with them post hoc. More systematic causal and intervention-based study designs — including controlled ablations that separate confidence calibration from raw uncertainty magnitude — would strengthen the evidentiary basis for uncertainty-driven mitigation strategies.

Unified epistemic-aleatoric decomposition for LLMs. While Malinin and Gales (2021) formalized this decomposition for structured prediction generally, its practical, efficient estimation for modern billion-parameter autoregressive LLMs — without resorting to expensive ensembles — remains an open problem. Techniques that can cheaply separate “the model doesn’t know the fact” from “the fact admits multiple valid phrasings” at the token level would substantially sharpen uncertainty-based hallucination signals.

Efficient semantic uncertainty at scale. Semantic entropy and its variants offer the strongest resolution of the lexical-confound problem but remain computationally expensive in their original sampling-based form. Further development of single-pass or low-sample approximations (in the spirit of semantic entropy probes) that retain semantic-level fidelity, and systematic benchmarking of the accuracy-efficiency trade-off across model scales and domains, is an important applied direction.

Standardized, cross-method benchmarks. The literature currently spans heterogeneous benchmarks, model families, and evaluation metrics (AUC-PR, AUROC, correlation with human factuality annotations, and others), making it difficult to compare token-level uncertainty methods against semantic, consistency-based, and internal-state methods on equal footing. Community benchmarks that jointly evaluate detection accuracy, computational cost, and applicability in black-box versus white-box settings would clarify the relative maturity of competing approaches.

Incentive-aware training and evaluation. Given the argument that current training and benchmark scoring structurally reward confident guessing over calibrated uncertainty expression (Kalai et al., 2025), an important direction is the joint redesign of training objectives and evaluation metrics so that token-level (and downstream verbal) uncertainty better tracks actual correctness — for instance, evaluation protocols that explicitly reward appropriate abstention and penalize confident hallucination more than honest uncertainty, potentially in tandem with reinforcement learning objectives that directly optimize calibration alongside helpfulness.

Uncertainty-aware decoding and retrieval integration. Further integration of token-level uncertainty signals into the decoding loop itself — extending beyond beam search modifications and attention-guided context decoding — including uncertainty-triggered retrieval augmentation (querying external knowledge sources specifically at high-uncertainty generation steps) represents a promising direction for combining the efficiency of parametric generation with the reliability of grounding in external evidence only where it is most needed.

Reasoning and agentic settings. As LLMs are increasingly deployed in multi-step reasoning and agentic pipelines where errors compound across steps and tool calls, extending token-level uncertainty methods to reliably flag unreliable intermediate reasoning steps — building on early work such as TokUR and TOKENHD — and studying how uncertainty propagates and compounds across a reasoning trajectory, is a critical open area with direct implications for the safety of autonomous LLM-based agents.

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7. Conclusion

Token-level uncertainty — spanning simple next-token entropy and log-probability through to sophisticated semantic and consistency-based extensions — has become one of the most productive lenses for anticipating hallucination in large language models. The empirical case for a meaningful relationship between predictive uncertainty and hallucination is well established across conditional generation, question answering, and machine translation settings (Xiao & Wang, 2021; Malinin & Gales, 2021; Farquhar et al., 2024), and this relationship has been operationalized in an increasingly diverse ecosystem of detection, fact-checking, and abstention systems. At the same time, the field has moved past treating raw token probability as a sufficient signal on its own: the recognized confounds between epistemic and aleatoric uncertainty, the mismatch between lexical variability and semantic (dis)agreement, and the documented tendency of modern instruction-tuned LLMs toward overconfidence collectively motivate the shift toward semantic-level, hybrid, and internal-state-informed uncertainty measures. Perhaps the most consequential open insight is that hallucination is not purely a matter of insufficient knowledge revealed through high uncertainty; it is also a structural consequence of training and evaluation incentives that reward confident guessing (Kalai et al., 2025). Future progress on using token-level uncertainty as a hallucination predictor will therefore likely depend not only on more refined uncertainty estimators, but on aligning the incentives under which LLMs are trained and evaluated with the goal of calibrated, honestly uncertain generation.

References

1. Azaria, A., & Mitchell, T. (2023). *The internal state of an LLM knows when it's lying*. arXiv preprint. arXiv:2304.13734. https://arxiv.org/abs/2304.13734

2. Chen, C., Liu, K., Chen, Z., Gu, Y., Wu, Y., Tao, M., Fu, Z., & Ye, J. (2024). *INSIDE: LLMs' internal states retain the power of hallucination detection*. arXiv preprint. arXiv:2402.03744. https://arxiv.org/abs/2402.03744

3. Duan, J., Cheng, H., Wang, S., Zavalny, A., Wang, C., Xu, R., Kailkhura, B., & Xu, K. (2024). Shifting attention to relevance: Towards the predictive uncertainty quantification of free-form large language models. arXiv preprint. arXiv:2307.01379. https://arxiv.org/abs/2307.01379

4. Fadeeva, E., Rubashevskii, A., Shelmanov, A., Petrakov, S., Li, H., Mubarak, H., Tsymbalov, E., Kuzmin, G., Panchenko, A., Baldwin, T., Nakov, P., & Panov, M. (2024). Fact-checking the output of large language models via token-level uncertainty quantification. In *Findings of the Association for Computational Linguistics: ACL 2024* (pp. 9367–9385). Association for Computational Linguistics. https://aclanthology.org/2024.findings-acl.558/

5. Farquhar, S., Kossen, J., Kuhn, L., & Gal, Y. (2024). Detecting hallucinations in large language models using semantic entropy. *Nature*, 630(8017), 625–630. https://doi.org/10.1038/s41586-024-07421-0

6. Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. (2017). On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning* (PMLR vol. 70, pp. 1321–1330). https://proceedings.mlr.press/v70/guo17a.html

7. Huang, L., Yu, W., Ma, W., Zhong, W., Feng, Z., Wang, H., Chen, Q., Peng, W., Feng, X., Qin, B., et al. (2025). A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *ACM Transactions on Information Systems*, 43(2), 1–55. https://doi.org/10.1145/3703155 (see also arXiv:2311.05232)

8. Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y. J., Madotto, A., & Fung, P. (2023). Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12), Article 248, 1–38. https://doi.org/10.1145/3571730 (preprint: arXiv:2202.03629)

9. Kadavath, S., Conerly, T., Askell, A., Henighan, T., Drain, D., Perez, E., Schiefer, N., Hatfield-Dodds, Z., DasSarma, N., Tran-Johnson, E., Johnston, S., El-Showk, S., Jones, A., Elhage, N., Hume, T., Chen, A., Bai, Y., Bowman, S., Fort, S., Ganguli, D., Hernandez, D., Jacobson, J., Kernion, J., Kravec, S., Lovitt, L., Ndousse, K., Olsson, C., Ringer, S., Amodei, D., Brown, T., Clark, J., Joseph, N., Mann, B., McCandlish, S., Olah, C., & Kaplan, J. (2022). *Language models (mostly) know what they know*. arXiv preprint. arXiv:2207.05221. https://doi.org/10.48550/arXiv.2207.05221

10. Kalai, A. T., Nachum, O., Vempala, S. S., & Zhang, E. (2025). *Why language models hallucinate*. arXiv preprint. arXiv:2509.04664. https://arxiv.org/abs/2509.04664

11. Kossen, J., Han, J., Razzak, M., Schut, L., Malik, S., & Gal, Y. (2024). *Semantic entropy probes: Robust and cheap hallucination detection in LLMs*. arXiv preprint. arXiv:2406.15927. https://arxiv.org/abs/2406.15927

12. Kuhn, L., Gal, Y., & Farquhar, S. (2023). Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In *The Eleventh International Conference on Learning Representations (ICLR 2023)*. https://openreview.net/forum?id=VD-AYtP0dve (preprint: arXiv:2302.09664)

13. Malinin, A., & Gales, M. (2021). Uncertainty estimation in autoregressive structured prediction. In *9th International Conference on Learning Representations (ICLR 2021)*. https://openreview.net/forum?id=jN5y-zb5Q7m (preprint: arXiv:2002.07650)

14. Manakul, P., Liusie, A., & Gales, M. (2023). SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing* (pp. 9004–9017). Association for Computational Linguistics. https://aclanthology.org/2023.emnlp-main.557/ (preprint: arXiv:2303.08896)

15. Min, R., Pang, T., Du, C., Cheng, M., & Fung, Y. R. (2026). *Scalable token-level hallucination detection in large language models* (TOKENHD). arXiv preprint. arXiv:2605.12384. https://arxiv.org/abs/2605.12384

16. Xiao, Y., & Wang, W. Y. (2021). On hallucination and predictive uncertainty in conditional language generation. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume* (pp. 2734–2744). Association for Computational Linguistics. https://doi.org/10.18653/v1/2021.eacl-main.236 (preprint: arXiv:2103.15025)

17. Zhang, T., Qiu, L., Guo, Q., Deng, C., Zhang, Y., Zhang, Z., Zhou, C., Wang, X., & Fu, L. (2023). *Enhancing uncertainty-based hallucination detection with stronger focus*. arXiv preprint. arXiv:2311.13230. https://arxiv.org/abs/2311.13230

18. *Detecting token-level hallucinations using variance signals: A reference-free approach*. (2025). arXiv preprint. arXiv:2507.04137. https://arxiv.org/abs/2507.04137

19. *Dynamic attention-guided context decoding for mitigating context faithfulness hallucinations in large language models*. (2025). arXiv preprint. arXiv:2501.01059. https://arxiv.org/abs/2501.01059

20. *TokUR: Token-level uncertainty estimation for large language model reasoning*. (2025). arXiv preprint. arXiv:2505.11737. https://arxiv.org/abs/2505.11737

21. *Token-level density-based uncertainty quantification methods for eliciting truthfulness of large language models*. (2025). arXiv preprint. arXiv:2502.14427. https://arxiv.org/abs/2502.14427

Note on sourcing: This review draws on peer-reviewed publications (e.g., Farquhar et al., 2024, published in Nature; Ji et al., 2023, published in ACM Computing Surveys; Manakul et al., 2023 and Xiao & Wang, 2021, both published at ACL-affiliated venues; Guo et al., 2017, published at ICML) as well as arXiv preprints reflecting recent and ongoing research (some not yet peer-reviewed at the time of writing). Readers should consult venue and peer-review status, indicated above, when weighing the evidentiary strength of individual claims.